

ALEX MONTELLO

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Education

Purdue University – B.S. in Mechanical Engineering
GPA – 3.97/4.00

West Lafayette, IN
Expected Graduation: May 2029

Experience

Purdue Formula SAE

West Lafayette, IN

Suspension Lead

May 2026 – Present

- Lead a team of 16 through design and manufacturing of uprights, wheel assemblies, actuation, and brakes built on NASA standard FoS, hand analysis, FEM, and exemplary design justification
- Coordinate 5+ research and validation projects such as weld testing, strain gauges, and composite control arms
- Design lighter and stiffer rotating wheel assembly, developing bolted joint calculator and transient thermal model

Chassis Subteam Member

Aug 2025 – May 2026

- CNC Machined 35+ chassis and dyno parts to specification and produced drawings with GD&T to ASME Y14.5
- Designed and implemented a mechanical throttle mechanism to ensure reliability and rule compliance
- Assembled and tested a pneumatic shifting system, reduced shift times by 76% by adding a plenum
- 2026 vehicle achieved 1st overall in North America, 3rd in Design with parts I manufactured and analyzed

Super Systems, Inc.

Cincinnati, OH

CAD/Systems Intern

May 2026 – Aug 2026

- Created detailed models and assembly drawings for 2 gas flow panels to support purchasing and assembly
- Programmed 3 automation tools that reduced multi-hour wiring and mechanical drafting tasks to seconds
- Completed installation updates on 30+ furnace electrical and mechanical drawings, finishing project lifecycles
- Wired and tested SCADA equipment for 5 heat treating furnaces with multimeter and signal generator

FIRST Tech Challenge Robotics

Cincinnati, OH

Mechanical Lead

Aug 2022 – June 2025

- Designed 100+ robot components prioritizing DFMA, parallel mechanisms, and optimized packaging
- Led a team of 15 through iterations using CAD version control, vision based analysis, and KPI tracking in Excel
- Won 8 awards across design, control, and engineering process, including at the international level

Projects

Articulated Robotic Arm

- Created a compact multi-DOF arm with worm-driven joints to prevent backdriving and centralize motor mass
- Integrated a compliant differential wrist claw enabling robust, forgiving grasping from multiple orientations

Multilayer Rubik's Cube Solving Robot

- Developed a novel machine with 6 modular grippers allowing decoupled linear and rotational motion for speed
- Achieved 1.1 second move cycles and eliminated risk of gripper crashes using compliant materials and software

Cable Driven Parallel Robot

- Designed and built a six-cable parallel manipulator driven by six servo-actuated pulley arms to maximize travel
- Achieved 0.9 m/s and 7.5 rad/s linear and angular speeds moving an effector around a 75 L workspace

Skills

- **CAD/CAM:** Solidworks, Onshape, Autodesk Inventor, AutoCAD, Siemens NX/Teamcenter, Fusion 360 CAM
- **Manufacturing:** CNC mills, manual mills, manual lathes, CNC lasers, electronics
- **Analysis:** Ansys Mechanical, MATLAB, Excel
- **Programming:** Python, Java, C, C#